Time left 1:08:55 Hide

Information

Answer All Questions

Information

6.(A)

Question **1**

Not yet answered

Marked out of 13.00

Define data abstraction in DBMS. Evaluate the different levels of abstraction (view level, logical level, and physical level) with examples.

Edit View Insert Format Tools

p tiny

Information

(OR)

Information

6.(B)

Question **2**

Not yet answered

Marked out of 13.00

Analyze the different categories of data models in DBMS. Compare and contrast each type with appropriate real-world examples.

Edit View Insert Format Tools

p tiny

Information

7.(A)

Question **3**

Not yet answered

Marked out of 13.00

Explain Functional Dependency and its types with suitable examples. Also, discuss its role in database normalization.

Edit View Insert Format Tools

p tiny

Information

(OR)

Information

7.(B)

Question **4**

Not yet answered

Marked out of 13.00

WWhat are the different types of anomalies in a database? How does normalization help in reducing these anomalies? Provide examples.

Edit View Insert Format Tools

p

tiny

Information

8.(A)

Question **5**

Not yet answered

Marked out of 14.00

Design an Entity-Relationship (ER) diagram for a University Management System. Identify the key entities, attributes, relationships, and constraints. Convert the ER diagram into relational tables and explain the process in detail.

Edit View Insert Format Tools

p

tiny

Information

(OR)

Information

8.(B)

Question **6**

Not yet answered

Marked out of 14.00

An institute maintains details of students, the courses they enroll in, and the faculty teaching those courses. A sample of the unnormalized data is shown below:

StudentID	StudentName	Courses (CourseID, CourseName, FacultyName, FacultyPhone)
S101	Priya	{(C01, DBMS, Dr. Kumar, 9876543210), (C03, AI, Dr. Ramesh, 9876545678)}
S102	Arjun	{(C02, Networks, Dr. Mehta, 9876501234)}
S103	Kavya	{(C01, DBMS, Dr. Kumar, 9876543210)}

(i) Explain the insertion, deletion, and update anomalies present in the above table. (5)

(ii) Normalize the table up to First Normal Form (1NF). (3)

(iii) Further normalize the table to Second Normal Form (2NF) and Third Normal Form (3NF). Clearly state any assumptions made and show the resulting relations. (7)

Edit View Insert Format Tools

p

tiny

◄ PART A - CIA 1

Jump to...

PART B and C ANSWER SCRIPT - CIA 1 ►